

Information Retrieval

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Information Retrieval

- Teachers: [Franco Maria Nardini](#) and [Rossano Venturini](#)
- CFU: 6
- Period: First semester
- Language: English
- Telegram group: [here](#)
- Teams group: [here](#)
- Lectures schedule: Monday 16:00-18:00 (Aula Fib C) and Wednesday 11:00-13:00 (Aula Fib L1)
- Question time: After lectures or by appointment. The fastest way to contact us is by email or Telegram. Don't be shy!

Bonus lectures: two guests from famous IR industries at the end of the course



Telegram



What is Information Retrieval?

Let's interview a IR system to present the topic and motivate our interest in IR!



Dear chatGPT, are you an information retrieval system?

Yes, in a way, I can be considered an advanced **information retrieval (IR) system**, but with added capabilities. Here's how I compare to traditional IR systems:

Similarities to IR Systems:

1. **Information Retrieval:** Like IR systems, I retrieve and present relevant information based on a user's query. I search through vast amounts of knowledge, identify relevant information, and respond in a contextually appropriate manner.
2. **Natural Language Processing (NLP):** I use advanced NLP techniques to understand the context of user queries, similar to how modern IR systems interpret and refine search results based on natural language queries.
3. **Ranking and Relevance:** I prioritize and synthesize information that I "retrieve" based on relevance, much like search engines use algorithms to rank results by relevance to a user's query.

What is Information Retrieval?

Let's interview a IR system to present the topic and motivate our interest in IR!



Dear chatGPT, are you an information retrieval system? (short answer, please. we just want to motivate our students to study)



Yes, I am an advanced information retrieval system, but I also generate new information by understanding queries, retrieving relevant data, and creating responses—combining the strengths of search engines with conversational AI!

Information Retrieval

What is Information Retrieval?

Information Retrieval (IR) is the process of obtaining relevant information from a large repository of unstructured data, such as documents, web pages, or multimedia content, based on a user's query. It involves searching and indexing data to find the most relevant pieces of information, and it's a fundamental component of technologies like search engines, recommendation systems, and digital assistants.

Key Components of IR:

1. **Document Representation:** IR systems process and represent data (e.g., text, images) in a structured way to make searching efficient. This typically involves tokenization, parsing, and indexing.
2. **Query Processing:** When a user inputs a query, the system interprets it, often breaking it down into keywords or semantic elements to match it with indexed data.
3. **Ranking and Relevance:** The core of IR is determining which documents or data are most relevant to the user's query. This involves ranking algorithms that score the relevance of each result based on factors like keyword frequency, semantic matching, or user behavior.
4. **Evaluation Metrics:** Common metrics to assess IR system performance include precision (how relevant the returned results are) and recall (how many relevant documents the system was able to retrieve).

Who still uses a search engine?

how is chatGPT able to answer to question that requires fresh information which are unknown at training time?

Worked for 45s >

ChatGPT can answer questions about fresh information because it **can use information supplied at answer time**, as well as knowledge learned during training.

There are two relevant sources of information:

- **Training:** builds the model's language skills, general knowledge, and ability to reason. This knowledge has a cutoff.
- **Context:** contains your messages and any documents or tool results provided during the conversation. This can include information published moments ago.

For example, if you ask, "Who won yesterday's match?", a search-enabled system can:

1. Request a web search for the match.
2. Receive recent results and read relevant pages.
3. Add that retrieved information to the model's context.
4. Generate an answer based on those sources, with citations. OpenAI documents this ability to retrieve up-to-date information through its  [web search tool](#).

This does not require retraining the model or changing its weights. It is similar to someone reading a new newspaper article and then answering questions about it using their existing reading and reasoning skills. You can achieve the same basic effect by pasting a new article into the conversation.

This general approach is often called **retrieval-augmented generation (RAG)**: retrieve relevant information, then use it to generate an answer.

Who still uses a search engine?

what's agentic search?

Agentic search is search where an AI decides what to look for next based on what it has already found. It can run several searches, inspect sources, identify missing information, and refine its approach before answering.

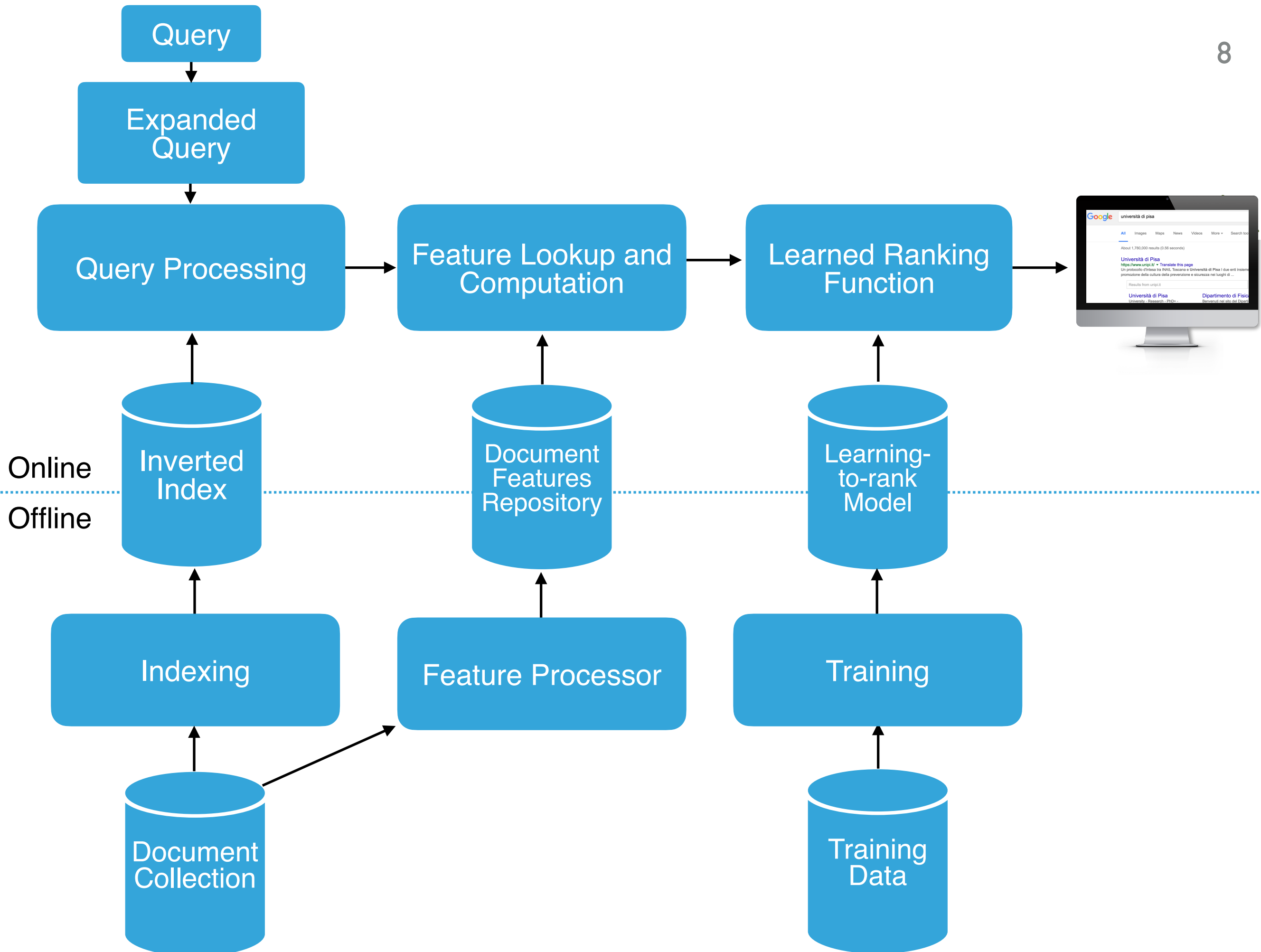
For example, you ask: "Why did Company X's profits fall last quarter?"

The AI might:

1. Find the latest earnings report.
2. Notice that profits fell despite higher revenue.
3. Look for explanations of rising costs.
4. Read the relevant report sections and search for supporting evidence.
5. Combine its findings into an answer with citations.

The key difference is **an adaptive feedback loop**:

Search → inspect results → identify gaps → search again → answer



Information Retrieval

Give me a list of 10 big companies that uses IR systems for their main business



Here are 10 large companies that use Information Retrieval (IR) systems as a core part of their main business:

1. **Google** – The world's largest search engine is built on IR, serving billions of queries daily.
2. **Microsoft (Bing)** – Microsoft's Bing search engine is a major competitor in the search space, powered by IR.
3. **Yahoo** – Though smaller today, Yahoo's search engine and related services are still IR-based.
4. **Amazon** – Product search and recommendations rely heavily on sophisticated IR systems.
5. **Facebook (Meta)** – IR is critical for social media content discovery and personalized feeds.
6. **Netflix** – Uses IR systems to power content recommendations and search for users.
7. **Spotify** – Music discovery and recommendations are driven by advanced IR technologies.
8. **Alibaba** – Its e-commerce platform uses IR to help users search for and discover products.
9. **eBay** – As an online marketplace, eBay's search and recommendation engines are powered by IR.
10. **Pinterest** – Visual search and content discovery are key aspects of Pinterest's business model, supported by IR systems.

These companies rely heavily on IR systems to drive their core business offerings, such as search, recommendations, and content discovery.

Let's play!

Name a missing company and say why.



Motivations

What's the entry-level salary in a big tech company for a software engineer?

The entry-level salary for a software engineer at a big tech company can vary based on location, company, and experience. However, typical entry-level salaries at major tech companies in the U.S. often range from **\$100,000 to \$150,000** per year, with some variation. Here's a breakdown:

1. Google:

- **Base Salary:** Around \$120,000 to \$130,000
- **Total Compensation** (including bonuses, stock options): \$150,000 to \$180,000

2. Microsoft:

- **Base Salary:** Around \$110,000 to \$120,000
- **Total Compensation:** \$130,000 to \$160,000

3. Facebook (Meta):

- **Base Salary:** \$120,000 to \$130,000
- **Total Compensation:** \$150,000 to \$180,000

4. Amazon:

- **Base Salary:** Around \$105,000 to \$115,000
- **Total Compensation:** \$130,000 to \$160,000

Are tech companies allowing remote working?



Yes,

Motivations

What do you want to do?

Locations ▾

Experience ▾

Skills and qualifications ▴

Computer programming, finance degree, et

Degree ▴

☐ Pursuing degree

☐ Associate

☒ Bachelors

☐ Masters

☐ Ph.D.

Job types ▴

☐ Full-time

☐ Part-time

☐ Temporary

☒ Intern

Student Researcher, BS/MS, Fall 2024

Google

Mountain View, CA, USA; Ann Arbor, MI, USA; +26 more

Intern and Apprentice

Minimum qualifications

- Currently enrolled in a Bachelor's or Master's degree in Computer Science, Linguistics, Statistics, Biostatistics, Applied Mathematics, Operations Research, Economics, Natural Sciences, or related technical field.
- Experience in one area of computer science (e.g., Natural Language Understanding, Computer Vision, Machine Learning, Deep Learning, Algorithmic Foundations of Optimization, Quantum Information Science, Data Science, Software Engineering, or similar areas).

Learn more

Student Researcher, BS/MS, Fall 2024

Google

Waterloo, ON, Canada; Montreal, QC, Canada; +1 more

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Learn more

Is there any course at University of Pisa that helps for coding interviews at major tech companies?

Worked for 24s >

Yes—🌐 Competitive
goals explicitly include

See you today at 14:00 - Room H!

atch. Its stated
companies.

IR Systems Have Two Goals

Effectiveness



Efficiency



First Goal: Effectiveness

13



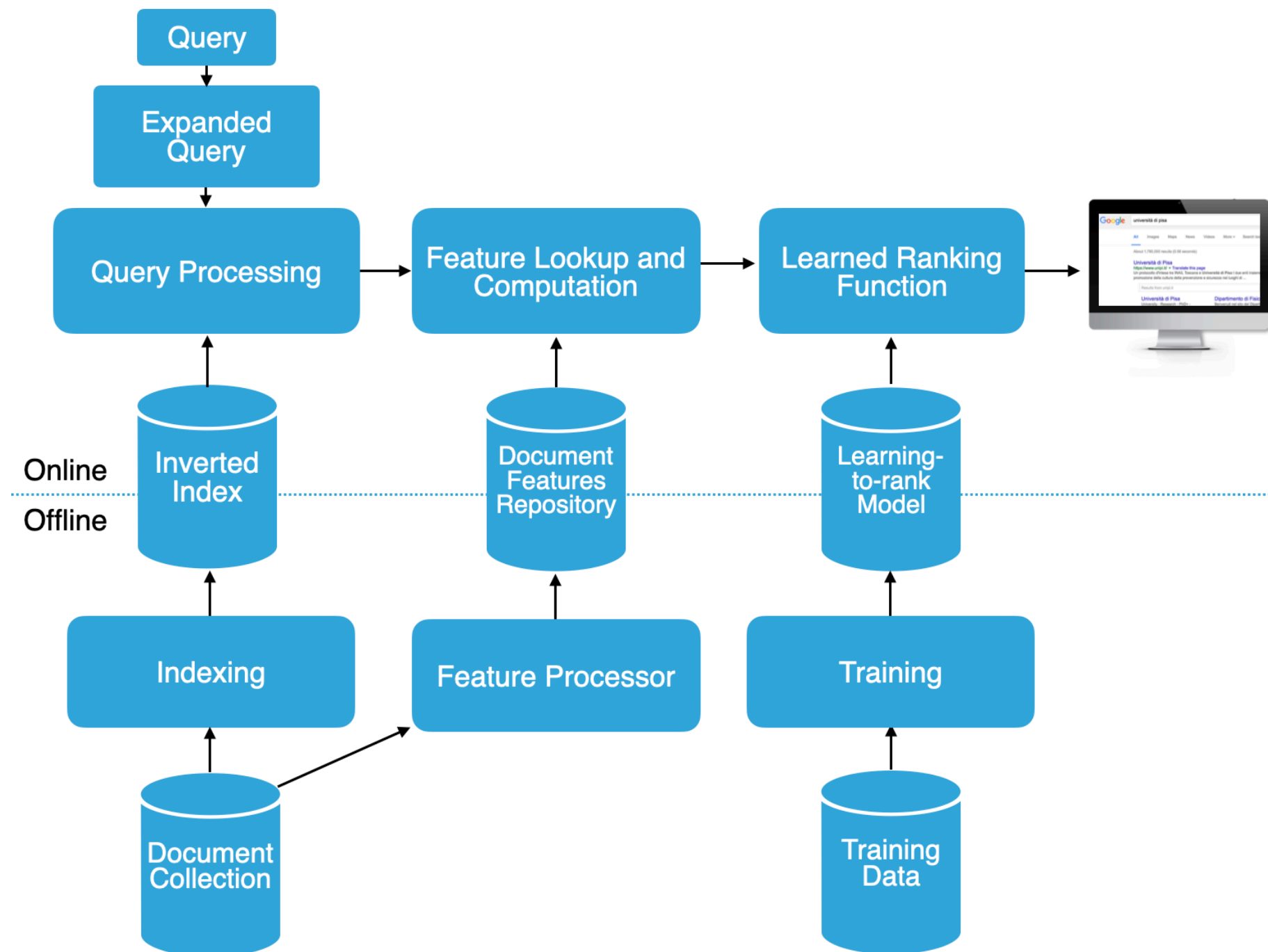
information retrieval unipi



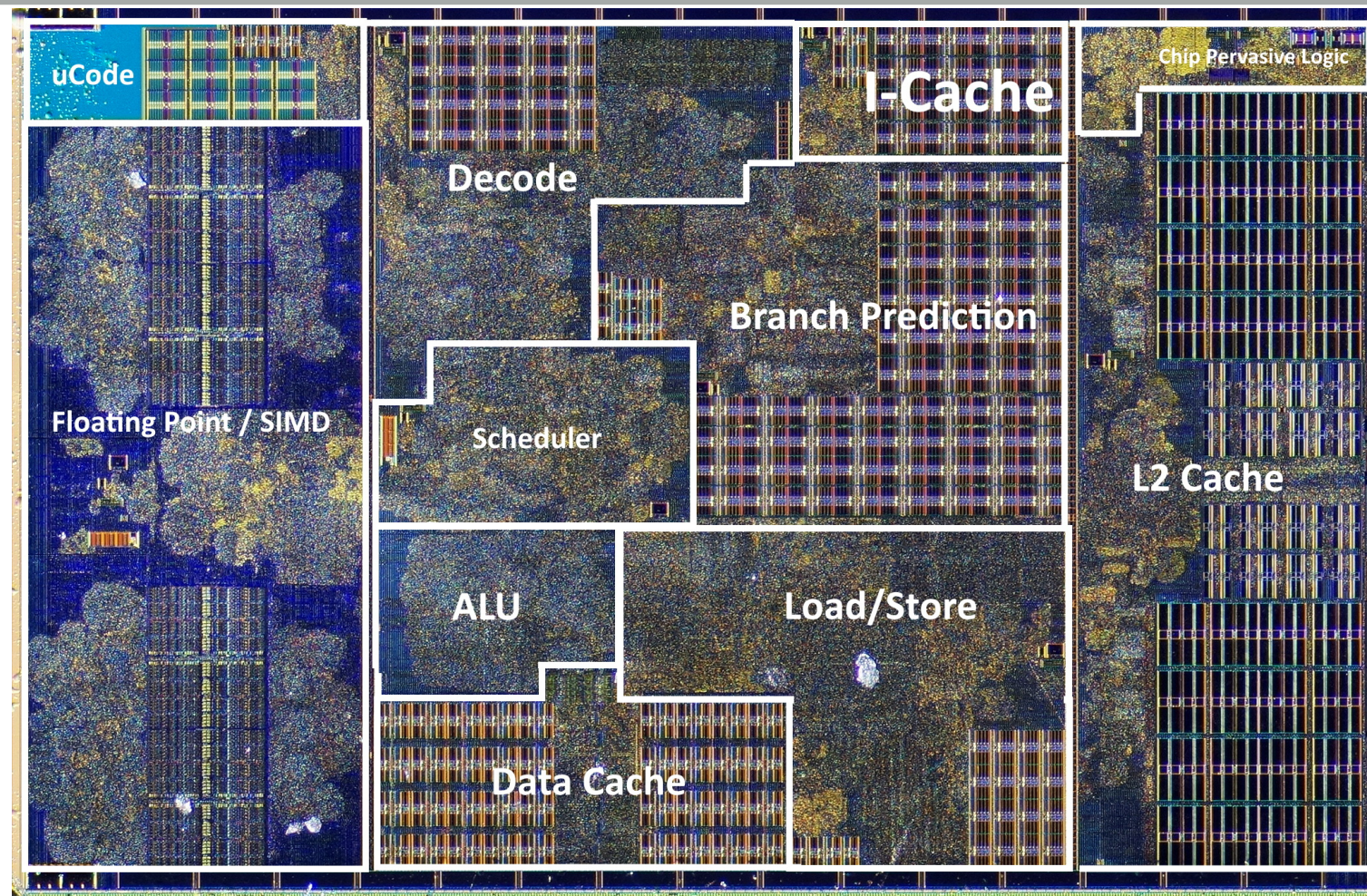
Google Search

I'm Feeling Lucky

Google offered in: [Italiano](#)



Second Goal: Efficiency



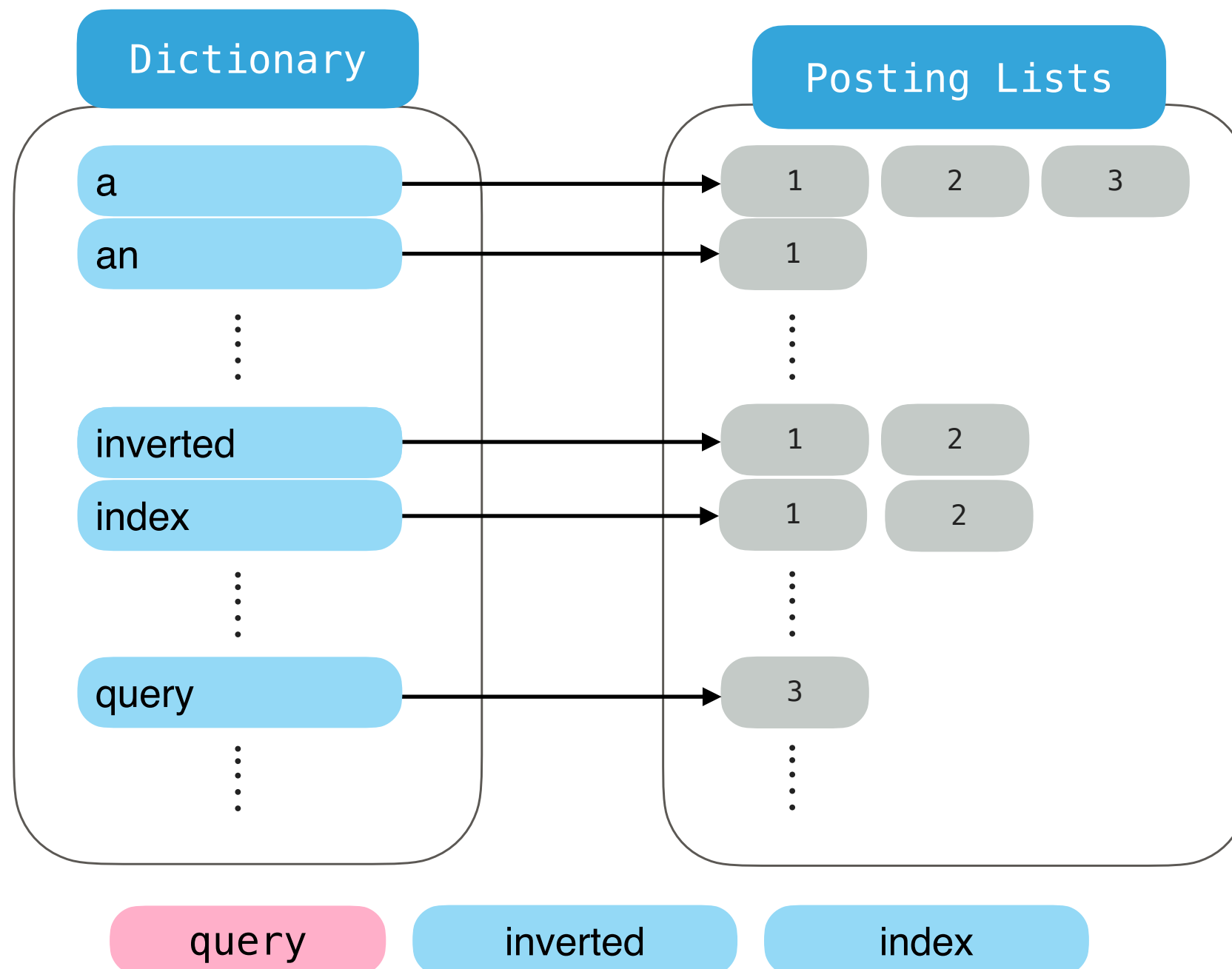
- Pipelining and Superscalar processors
- Pipeline Hazards
 - Structural, Data, and Control Hazards
- Memory Hierarchy
- SIMD
- Parallel Computing

Language Properties and IR Models

- What are the properties of the language?
 - ▶ How do they affect retrieval?
- What does it mean to “retrieve” a piece of text?
- Document representations
- Retrieval models
 - ▶ Boolean
 - ▶ TF-IDF
 - ▶ BM25 (and variations)

$$score(q, d) = \sum_{i=1}^{|q|} idf(q_i) \cdot \frac{tf(q_i, d) \cdot (k_1 + 1)}{tf(q_i, d) + k_1 \cdot (1 - b + b \cdot \frac{|d|}{avgdl})}$$

- d_0 An inverted index is an index storing a mapping from terms to their occurrences in a set of documents.
- d_1 The inverted index data structure is a central component of a typical search engine indexing algorithm.
- d_2 The goal is to optimize the speed of the query: find the documents where a set of words occur.





Advanced Search

Find pages with...

all these words:

| Conjunctive AND

this exact word or phrase:

Phrase Search

any of these words:

information retrieval Disjunctive OR

none of these words:

NOT

numbers ranging from:

to

To do this in the search box.

Type the important words: tri-colour rat terrier

Put exact words in quotes: "rat terrier"

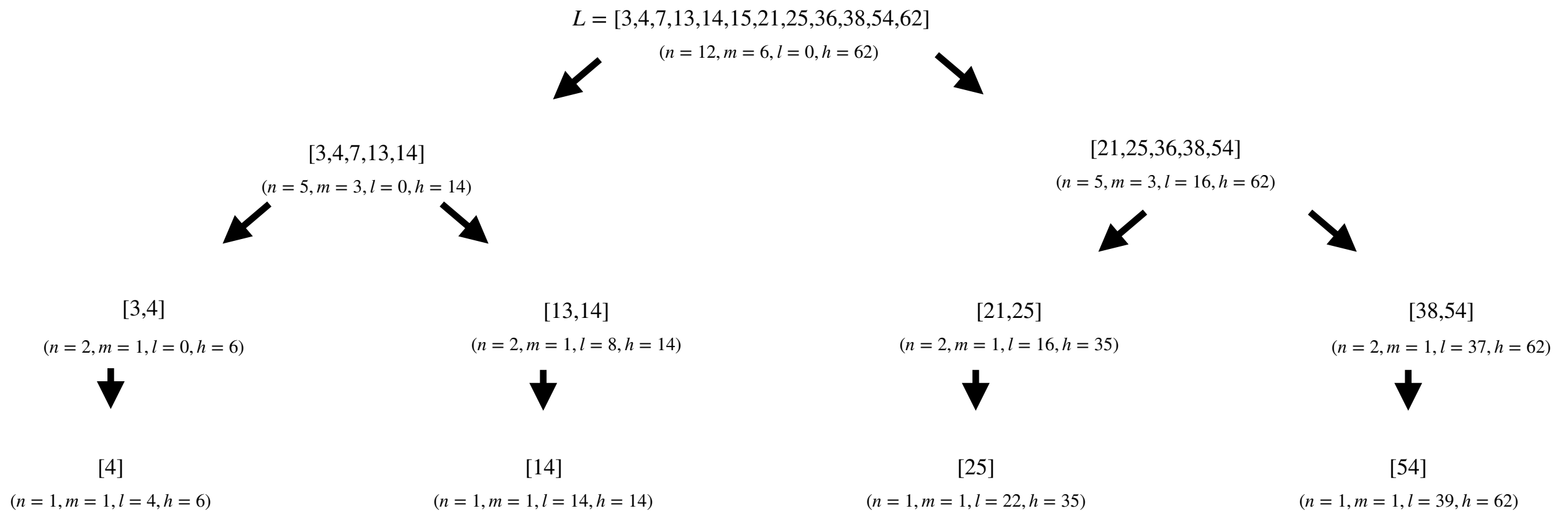
Type OR between all the words you want: miniature OR standard

Put a minus sign just before words that you don't want:
-rodent, -"Jack Russell"

Put two full stops between the numbers and add a unit of measurement:
10..35 kg, £300..£500, 2010..2011

Data Compression

Binary Interpolative Coding



Lempel-Ziv '77

T a b a b a a a a b a b a a a a b

$(0, a) (0, b) (2, 3) (1, 3) (7, 8)$

gzip

7ZIP

LZ4

Learning-to-Rank Models

large training set of queries and ideal document ranking

$q_a, \langle (d_1, y_{a1}), (d_2, y_{a2}), (d_3, y_{a3}), (d_5, y_{a5}), (d_8, y_{a8}), (d_{13}, y_{a13}), \dots \rangle$

$q_b, \langle (d_{99}, y_{b99}), (d_{98}, y_{b98}), (d_{97}, y_{b97}), (d_{96}, y_{b96}), (d_{95}, y_{b95}), \dots \rangle$

Machine Learning
(a.k.a. Black Magic)

Example of label y :
five values, from
0 (irrelevant) to
4 (perfectly relevant)

query,
 $d_1, d_2, d_3, \dots$

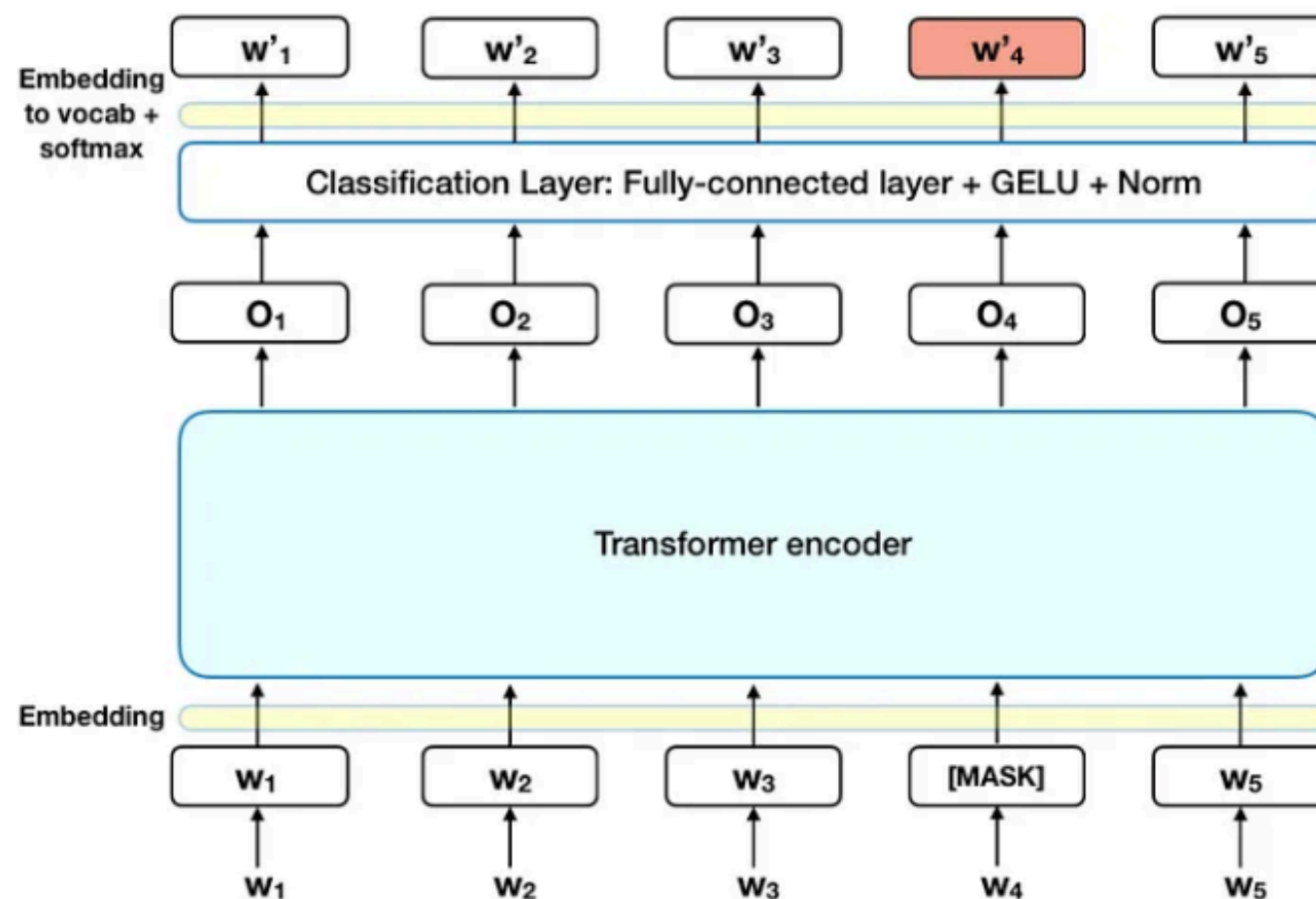
Black Box

ranked list
 $d_{17}, d_{13}, d_{666}, \dots$

- ❑ The goal is to learn the *ranking*, not the label
- ❑ **Not so easy**: sorting is not nicely derivable
 - ❑ Cannot apply *gradient descent* or similar...

Neural Information Retrieval

- Deep Learning heavily impacts IR as many other research areas
- From 2018, BERT (transformer models)
- Two different lines of research
 - Representation-based techniques
 - Dense and Sparse models
 - Interaction-based techniques, i.e., Cross-encoders



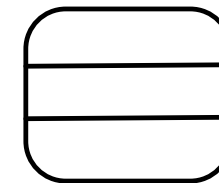
Neural Information Retrieval

- Large language models encode documents and queries into **embeddings**
- Three different families of embeddings
 - Dense **single-vector**
 - Dense **multi-vector**
 - **Sparse single-vector**
- Sparse embeddings are
 - Highly **effective** for in-domain and out-of-domain evaluation
 - More space and time efficient than dense multi-vector
 - **Interpretable** by design

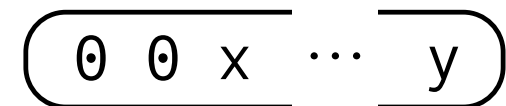
Single
Vector



Multi
Vector



Sparse
Vector



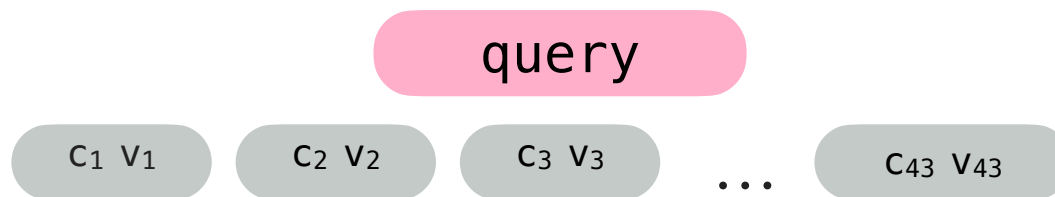
Query:

'what shoes do most nba players wear'

Representation:

nba: 2.56
shoes: 2.32
shoe: 2.14
basketball: 1.97
wearing: 1.55
wear: 1.52
most: 1.34
players: 1.25

K Approximate Nearest Neighbour Search

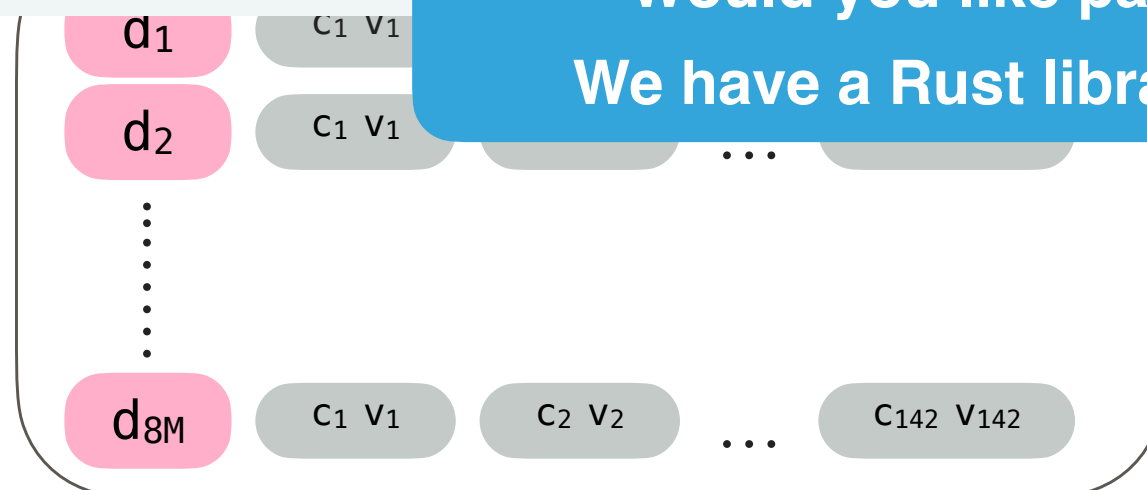


NeurIPS'23 Competition Track: Big-ANN

Supported by  Microsoft  Pinecone  aws  zilliz

New: the latest ongoing leaderboard has been released (March 1st, 2024).

Would you like participate?
We have a Rust library for KNN!



s: ~30K
non-zero components: ~150

TusKANNy (<https://tuskanny.com/>)



Building state-of-the-art algorithms and open-source tools for **Approximate Nearest Neighbor Search** across dense, sparse, and multi-vector representations.

A software ecosystem from Pisa, Tuscany, Italy — [ISTI-CNR](#) · [University of Pisa](#)



#	Team	Code	Paper	Algorithm	Recall	QPS	Build (s)	Query (s)
1	tuskanny-seismic	seismic_sisap2026	under review	seismic	0.9036 ✓	15,870	668.21	0.22
2	tuskanny-kannolo	kannolo_sisap2026_challenge	under review	kannolo-hnsw-rerank	0.9045 ✓	9,320	9017.07	0.37
3	hanswurst	sisap26challenge	under review	HaNSWurst	0.9005 ✓	958	2135.42	3.60
4	basque-invaders	nmf-ivf	under review	nmf-ivf	0.9048 ✓	404	37883.42	8.54

Exam Structure Overview

- **Part 1:** Written Exam (up to 28 points) with open answers
 - 1.5 hours written exam covering topics discussed during the course.
 - Questions will assess the understanding of the core course materials.
 - Mandatory to pass the exam with 18+.
- **Part 2 (optional):** Presentation of the results of a project (up to 4 points)
 - The project is based on a scientific paper from over 15+ selected papers.
 - The project will be presented using approximately 10 slides (15 min.) during the oral exam. Questions on experimental methodology and source code used.
 - Students can approach the project at different levels of depth:
 - **Basic Level:** Read and understand the techniques presented in the article **plus** conduct experiments to verify **parts of** the assumptions, main properties, and conclusions.
 - **Intermediate Level:** Replicate the results from the article.
 - **Advanced Level:** Re-implement the proposed solution, and optionally improve it by introducing alternative strategies.
 - **Projects can become a final thesis! Ask us!!!**

Coding with LLM

AI Coding Tools — University Student Pricing in Italy

Official euro prices, without currency conversion. Checked on 15 September 2026.

Tool	Free access for students	Paid pricing	Coding features
 GitHub Copilot	€0 with Copilot Student, after GitHub Education verification.	No paid upgrade required for the student plan.	In-editor code completion, chat, and coding agents. AI usage is limited; models are selected automatically.
 Gemini	Google AI Plus: €0 for 12 months for eligible university students aged 18+.	€4.99/month afterwards unless cancelled. Optional  AI Pro : €21.99/month.	Generates, explains, and debugs code in chat. Google also offers a separate free coding-agent tier through Antigravity.
 OpenAI Codex	€0 on ChatGPT Free, with limited usage. No Italy-specific student promotion verified.	ChatGPT Go: €8/month, Plus: €23/month, Pro: from €103/month.	Edits project files, runs commands and tests, and works through the app, terminal, and IDE extension.
 Claude / Claude Code	€0 for Claude Free chat. Claude Code is not included; no universal individual student offer verified.	Pro: €18/month + VAT, or €180/year + VAT. Includes Claude Code. Max starts at €90/month + VAT.	Chat-based coding assistance; Claude Code edits project files and runs commands directly.

Notes

- **Gemini:** claim by **31 December 2026**. Student verification and a payment method are required; renewal is automatic.

